

COLLAFIRTH HILL, United Kingdom

WMO: 030040

Lat: 60.5336N

Lon: 1.3910W

Elev: 228

StdP: 98.62

Time Zone: 0.00 (EUW)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.6	-2.6	-5.9	2.3	-1.9	-4.8	2.6	-1.3	25.9	2.1	23.3	3.6	8.8	30	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	16.5	13.4	14.5	12.2	13.2	11.6	13.9	16.0	12.9	13.9	12.0	12.7	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	9.6	14.3	12.3	9.1	13.2	11.6	8.8	12.4	39.6	15.9	36.9	14.0	34.9	12.7	17.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	20.6	18.2	DB	-5.2	19.3	0.7	2.0	-5.8	20.8	-6.2	21.9	-6.6	23.1	-7.1	24.6	(4)
(5)			WB	-5.7	15.8	0.7	1.4	-6.2	16.8	-6.6	17.7	-6.9	18.5	-7.4	19.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	2.4	2.0	2.3	3.8	6.7	8.7	10.7	10.7	8.5	7.1	4.4	3.4	(6)	
	DBStd	3.72	2.40	2.08	2.13	2.25	2.13	2.12	1.72	1.63	1.69	1.78	2.27	2.33	(7)	
	HDD10.0	1574	236	224	239	186	105	51	11	9	49	92	169	204	(8)	
	HDD18.3	4534	494	457	497	436	360	289	238	237	294	349	419	462	(9)	
	CDD10.0	82	0	0	0	0	3	11	31	31	5	1	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.9	10.7	9.9	10.4	8.1	7.7	6.8	7.2	7.4	9.4	9.8	9.6	10.2	(14)	
(15) Precipitation	PrecAvg	1154	145	113	106	64	59	57	60	82	91	115	130	133	(15)	
	PrecMax	1395	278	174	186	107	123	109	131	160	140	182	195	233	(16)	
	PrecMin	980	46	51	39	22	9	10	22	38	28	47	58	51	(17)	
	PrecStd	117	52	30	28	18	24	24	25	28	28	33	33	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	7.9	7.7	10.8	16.4	18.5	18.6	19.1	13.3	11.9	9.2	8.6	(19)	
		MCWB	7.9	7.1	7.2	8.0	11.1	14.1	15.2	14.8	12.0	11.2	8.9	8.2	(20)	
	2%	DB	7.4	6.8	7.0	8.9	13.3	15.6	16.2	15.4	12.3	10.7	8.6	8.0	(21)	
		MCWB	7.1	6.3	6.6	6.8	9.3	12.8	13.6	13.1	11.4	10.2	8.3	7.8	(22)	
	5%	DB	6.7	5.7	6.2	7.8	10.9	13.0	14.2	13.7	11.6	10.1	8.1	7.3	(23)	
		MCWB	6.2	5.0	5.7	6.7	8.1	10.8	12.5	12.3	10.9	9.7	7.8	7.0	(24)	
	10%	DB	5.7	5.0	5.5	7.0	9.5	11.4	13.0	12.8	11.0	9.5	7.6	6.7	(25)	
		MCWB	5.1	4.4	4.9	6.1	7.7	9.8	11.6	11.6	10.4	9.1	7.1	6.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.3	7.5	8.6	12.1	14.6	15.4	15.4	12.7	11.3	9.2	8.6	(27)	
		MCDB	8.1	7.5	7.6	10.1	15.7	17.7	17.8	17.7	12.7	11.5	9.2	8.7	(28)	
	2%	WB	7.2	6.2	6.6	7.6	9.9	12.8	14.0	13.7	11.7	10.4	8.5	7.9	(29)	
		MCDB	7.3	6.5	6.8	8.2	12.0	15.4	15.9	15.3	11.9	10.5	8.6	8.0	(30)	
	5%	WB	6.4	5.3	5.9	6.9	8.9	11.2	12.9	12.8	11.2	9.8	8.0	7.2	(31)	
		MCDB	6.6	5.7	6.1	7.5	10.4	12.4	13.8	13.4	11.4	10.0	8.1	7.3	(32)	
	10%	WB	5.3	4.5	5.1	6.2	8.0	10.3	12.0	11.9	10.6	9.2	7.3	6.4	(33)	
		MCDB	5.6	4.9	5.4	6.8	9.2	11.0	12.6	12.4	10.9	9.4	7.5	6.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	2.7	2.9	3.7	4.1	3.8	3.5	3.3	3.0	2.8	2.5	2.8	(35)	
		MCDBR	3.3	3.7	3.4	4.5	6.6	6.4	5.5	4.7	3.4	2.9	2.7	3.7	(36)	
		MCWBR	4.1	3.9	3.5	3.1	4.1	3.8	3.6	2.9	3.0	2.9	3.3	4.3	(37)	
	5% WB	MCDBR	3.3	3.6	3.3	4.0	5.8	5.6	4.9	4.3	3.1	3.0	2.8	3.6	(38)	
		MCWBR	4.2	4.1	3.8	3.4	4.1	3.8	3.8	3.1	3.2	3.2	3.4	4.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.317	0.342	0.356	0.369	0.359	0.353	0.366	0.353	0.351	0.335	0.311	0.413	(40)	
	taud		2.082	2.224	2.309	2.339	2.428	2.481	2.459	2.495	2.468	2.343	2.134	2.473	(41)	
	Ebn at Noon		392	629	763	826	867	878	855	839	771	639	398	263	(42)	
	Edh at Noon		46	71	89	103	101	98	98	88	76	62	42	31	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.22	0.71	1.79	3.34	4.66	4.84	4.36	3.54	2.14	1.07	0.32	0.14	(44)	
	RadStd		0.02	0.08	0.13	0.19	0.29	0.52	0.38	0.34	0.20	0.11	0.03	0.01	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		+0.34	+0.50	+0.72	N/A	N/A	N/A	-93	-123	N/A	N/A

Nomenclature: See separate page